Gas Mixtures and Gas Stoichiometry

Partial pressures, molar volume and gas volumes in reactions

Name _____ Class _____ Date _____

EQUATIONS AND DATA

$PV = nRT$ · $R = 0.08206$ L·atm/(mol·K) · $T(K) = t(°C) + 273.15$ · 1 atm = 760 mmHg = 101.325 kPa

$P(total) = P(A) + P(B) + \ldots$ · $P(A) = x(A)\,P(total)$ · molar volume 22.4 L/mol (0 °C, 1 atm)

Water vapour pressure: 19.8 mmHg at 22 °C · Molar masses (g/mol): He 4.003 · O_2 32.00 · Zn 65.38 · NaN_3 65.02

Show units in every step.

A. Partial pressures

1. State Dalton's law, and explain why the partial pressure of a gas is proportional to its mole fraction.

2. Dry air at 1.00 atm is 78.1 % N_2, 20.9 % O_2 and 0.93 % Ar by moles. Calculate each partial pressure in atm and in kPa.

3. 2.00 g of helium and 8.00 g of oxygen are placed in a 10.0 L container at 27 °C. Calculate (a) the moles of each gas (b) the mole fraction of helium (c) each partial pressure and the total pressure.

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B. Gases collected over water

4. Oxygen is collected over water at 22 °C. The volume is 0.500 L and the total pressure is 748 mmHg. Calculate (a) the pressure of the dry oxygen in atm (b) the moles and mass of oxygen collected.

C. Gas stoichiometry

5. 1.31 g of zinc reacts with excess hydrochloric acid: $\mathrm{Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2}$. Calculate the volume of hydrogen formed at 25 °C and 1.00 atm.

6. All gases are at the same temperature and pressure. (a) What volume of oxygen is needed to burn 2.00 L of propane? (b) What volume of CO_2 forms? $\mathrm{C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O}$

7. What volume does 0.250 mol of nitrogen occupy at 0 °C and 1 atm? Why can’t you use the same molar volume at 25 °C?

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D. Challenge

8. A car airbag inflates with nitrogen from the decomposition of sodium azide: $2\,NaN_3 \longrightarrow 2\,Na + 3\,N_2$. What mass of sodium azide is needed to fill a 60.0 L airbag at 25 °C and 1.10 atm?

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